

Publication Atlas, Caption Contract, and Scholarship Audit

Publication Atlas, Caption Contract, and Scholarship Audit I

The publication layer is generated from the same typed package state as the stimuli. It contains 15 scientific figures, 10 machine-readable tables, and a separately provenance-recorded editorial cover. The cover is an illustration of the project's subject and methods aesthetic; it is not a stimulus, a participant result, or evidence of a perceptual effect.

Claim-level visualization I

Claim-level boundary

Each stage exposes a different kind of evidence; later stages do not retroactively validate perception.

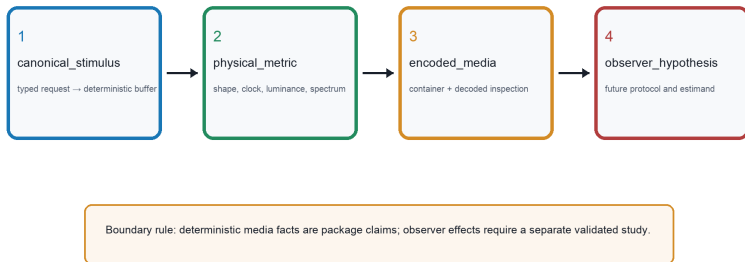


Figure 1: Four numbered boxes separate canonical stimulus, physical metric, encoded media, and observer hypothesis, with a boundary rule stating that observer effects require a separate study.

Claim-level visualization II

What this figure shows: Four numbered boxes separate canonical stimulus, physical metric, encoded media, and observer hypothesis, with a boundary rule stating that observer effects require a separate study. The four-stage boundary distinguishes what DuckRabbit can establish directly—canonical stimulus identity, physical/media metrics, and decoded encoded-media facts—from a future observer hypothesis. Source data: `output/data/claim_boundary.json` records the stage definitions and boundary rule; the arrows describe increasing evidential requirements, not an inference that one stage establishes the next. Controls: claim levels in the taxonomy and manifest; seed not applicable to the explanatory diagram. Objective facts: no unit-bearing objective quantity is applicable; stage definitions and evidence boundaries are preserved in the sidecar. Claim level: `source_supported`. Source data: `output/data/claim_boundary.json` (SHA-256 digest recorded in the figure registry). Evidence lineage: `formalism:eq:observer_estimand`, `formalism:eq:canonical_digest`. Limitations: The diagram is a documentation contract and does not replace a preregistered observer study or empirical data.

Scholarship map and lineage I

Scholarship: man and lineage II

Source-tiered scholarship map

18 catalog entries; 36 source records; exact claims, engineering bases, and limitations are retained in source data.

catalog entry	primary	review	theory	supported claim	status
visual_audiot_subst	1	1	0	The entry implements a controllable anisotropic figure-rotational family. Isolation recorded: The generation of the figure-rotational family is not feasible in practice.	implemented
visual_simultaneous_content	0	1	0	The physical center patches are nullified while sufficient luminance differs. Isolation recorded: The center patches and observer adaptation are not modeled.	implemented
visual_apparent_motion	1	1	0	The sequence contains a controlled successive spatial events. Isolation recorded: The spatial coordinates, temporal resolution, and perceived motion.	implemented
audio Shepard_tone	1	0	0	The audio contains octave-related patches with controlled sweep parameters. Isolation recorded: Physical phenomena and human perception are not modeled.	implemented
audio_missing_fundamental	0	1	0	The musical fundamental component is absent from the canonical spectrum. Isolation recorded: Pitch perception is an observer-dependent phenomenon.	implemented
audiovisual_sound_induced_flash	1	1	0	The declared brightness event occurs and offsets are physically encoded. Isolation recorded: The offset does not depend on stimulus size and contrast.	implemented
audiovisual_ventriloquist	0	2	0	The audio and visual channels carry a declared spatial discrepancy. Isolation recorded: Perceived location depends on vision, haptics, memory, and attention.	implemented
visual_ruler_bar	0	1	1	The two bar lengths are equal in the canonical space while being geometrically distinct. Isolation recorded: The chosen normalized geometry is not engineering-optimal.	implemented
visual_jogger_offset	0	1	1	The occluder and diagonal continuation are generated from explicit geometry. Isolation recorded: Alignment judgments and perception time are not measured.	implemented
visual_pencil	1	1	0	Target bars are physically equal and rate converge toward a matching region. Isolation recorded: Perceptual adaptation is observed and physically measured.	implemented
visual_kanizsa_triangle	1	1	0	The image contains incomplete inducers with no explicitly drawn figure edge. Isolation recorded: Motion contrast, completion, and depth are measured.	implemented
visual_ebbinghaus	2	0	0	Central target geometry is held equal while contextual size is geometrically differs. Isolation recorded: Perceived size depends on viewing angle and contrast.	implemented
visual_zollner	2	1	0	The entry implements a controlled crossing line orientation stimulus family. Isolation recorded: The raster analysis, the geometry, and the perceived orientation are not measured.	implemented
audio_tribase_pansonic	2	0	0	The canonical pair has an explicit half-octave frequency relation. Isolation recorded: Frequency perception is not measured and is not modeled.	implemented
audio_active_flicker	1	0	0	The two channels contain alternating octave-related tones. Isolation recorded: The auditory channel depends on stimulus presentation and observer.	implemented
audio_auditory_continuity	2	0	0	The canonical audio contains a reproducing interruption removal and random family. Isolation recorded: The generation of the stimulus depends on the physical observation and observer, not a feature of the stimulus or sensory channel adaptation.	implemented
audiovisual_temporal_ventriloquist	2	2	0	Audio and video event timing and declared offset are deterministic and perceptible. Isolation recorded: The generation of the stimulus is not a controlled engineering-optimal.	implemented
audiovisual_mcpuk	1	0	0	The catalog identifies a speech-dependent audiovisual family without claiming engineering bases. Isolation recorded: Technical, context, and observer-dependent effects are not modeled.	input_required

Formal traceability and objective metrics I

Formal traceability and objective metrics II

Formalism traceability

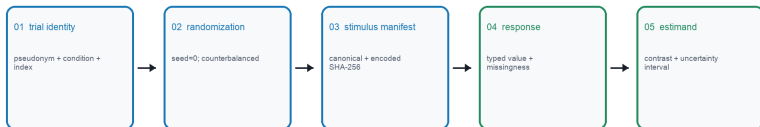
Equation labels resolve to implementation modules, tests, figures, and claim levels.

label	equation	implementation	tests	figures
eq_typed_request claim: canonical_stimulus	$q = \emptyset, 0, 5, e)$	<code>registry.py, schema.py</code>	<code>test_generators.py</code>	<code>fig_architecture,</code> <code>fig_formalism_traceability</code>
eq_canonical_generation claim: canonical_stimulus	$A = G(\theta, s)$	<code>registry.py, generators.py</code>	<code>test_generators.py,</code> <code>test_v03_contracts.py</code>	<code>fig_architecture,</code> <code>fig_visual_panel</code>
eq_canonical_digest claim: physical_metric	$c(A) = \text{Serialize_LE_float32}(A, \text{shape}, \text{clock}, \text{units});$ $h_c(A) = H(c(A))$	<code>canonical.py, manifest.py</code>	<code>test_v03_contracts.py</code>	<code>fig_architecture,</code> <code>fig_encoding_verification</code>
eq_encoding_verification claim: encoded_media	$F = E_d(A), I = D(F), V(F), M \sqsubseteq \{\text{press}, \text{tail}\}$	<code>render.py, inspection.py,</code> <code>manifest.py</code>	<code>test_media_and_render.py,</code> <code>test_v03_contracts.py</code>	<code>fig_encoding_verification</code>
eq_clock_definition claim: physical_metric	$N = \text{sound}(f_s T), L_k = k f_r \Delta, AV = L_{\text{audio}} - L_{\text{video}}$	<code>parameters.py, artifacts.py</code>	<code>test_parameters.py,</code> <code>test_media_and_render.py</code>	<code>fig_temporal_sequence,</code> <code>fig_audiovisual_timeline</code>
eq_objective_statistics claim: physical_metric	$\mu, Y, \sigma, Y, \text{RMS_X} = M(A, \text{units}, \text{tolerance}, \text{version})$	<code>metrics.py, publication.py</code>	<code>test_v04_scholarly.py</code>	<code>fig_metrics_dashboard,</code> <code>fig_audio_signals</code>
eq_temporal_spectral_metrics claim: physical_metric	$D_k = \text{mean absolute difference of adjacent frames};$ $\text{contrast}(X) = 2 \cdot \Phi_X(\text{ZP_X})$	<code>metrics.py, artifacts.py</code>	<code>test_v04_scholarly.py</code>	<code>fig_temporal_sequence,</code> <code>fig_metrics_dashboard</code>
eq_observer_estimand claim: observer_hypothesis	$\psi = E(Y \text{ conditional on condition and protocol})$	<code>observer_analysis.py, observer.py</code>	<code>test_v04_scholarly.py</code>	<code>fig_synthetic_psychophysics,</code> <code>fig_observer_protocol</code>
eq_synthetic_observer claim: synthetic_model_output	$p_M(c(A, r(A, c) = o)(w \psi(A, c) - w \psi(A, r))$	<code>synthetic_psychophysics.py,</code> <code>metrics.py</code>	<code>test_synthetic_psychophysics.py</code>	<code>fig_synthetic_psychophysics</code>

Observer-design boundary I

Observer protocol: orchestration boundary

The same stimulus manifest can feed a synthetic diagnostic now or a preregistered human study later; the data streams remain separate.



Two explicitly different endpoints

Synthetic psychophysics (included)

duckrabbit.synthetic.feature_observer
hand-specified features · analytic logistic rule
training_data = none · human_data = false
output: model diagnostic, not a psychometric function

Human psychophysics (future contract)

display / playback calibration participant
consent + pseudonymization preregistered
response, exclusions, model output: observer
estimate with uncertainty

Model templates for future human analysis

logistic_mixed: binomial / logit

lognormal_mixed: lognormal / identity

transition_rate: survival or transition-rate / hazard

linear_mixed: Gaussian / identity

poisson_or_ordinal: Poisson/negative-binomial/ordinal / log or cumulative

No participant records, fitted coefficients, or observer-level result is bundled; the included endpoint is model output only.

Observer-design boundary II

Figure 6: A flow diagram connects trial identity, randomization, stimulus manifest, response, aggregate estimand, and analysis-model templates, with a synthetic-only boundary.

Observer-design boundary III

What this figure shows: A flow diagram connects trial identity, randomization, stimulus manifest, response, aggregate estimand, and analysis-model templates, with a synthetic-only boundary. The study-ready scaffold moves from pseudonymous trial identity and deterministic randomization to a stimulus manifest and encoded-file hash, typed response or missingness, and a preregistered estimand. Source data output/data/observer_protocol.json records the study design, randomization seed, model templates, and synthetic-only status; no participant record is included. Controls: study design and deterministic randomization seed recorded in the sidecar; synthetic response generation is separate from human data. Objective facts: trial counts, condition structure, estimand templates, response schemas, and model families; participant outcomes are not applicable. Claim level: observer_hypothesis. Source data: output/data/observer_protocol.json (SHA-256 digest recorded in the figure registry). Evidence lineage: observer:preregistered_estimands. Limitations: The scaffold specifies future data collection and analysis but supplies no

Generated audit tables I

Table 1: Code-owned caption, source-data, and accessibility audit.

Figure / claim level	Source and evidence	Controls and objective facts	Limitations, boundary, and accessibility
fig:architecture; canonical stimulus	output/data/architecture.json formalism:eq:typed request, formalism:eq:canonical generation, formalism:eq:encoding verification	Controls: typed request q=(i, s,e) with seed s=0; default generator and delivery-independent canonicalization Objective: no unit-bearing objective quantity is applicable to this architecture diagram; stage identities and provenance fields are recorded in the sidecar	Limitations: The schematic abstracts implementation boundaries and does not represent an observer study or a causal theory of perception. Boundary: The pipeline verifies reproducible media facts, not what a person experiences. Accessibility: Stage names, mathematical symbols, and arrow direction are printed directly; color is redundant.
fig:catalog matrix; source supported	output/data/catalog matrix.json; gregory1997visual, hirst2020sound, bruns2019ventriloquist	Controls: live taxonomy registry; evidence matrix snapshot; seed not applicable to the catalog diagram Objective: 18 catalog rows; categorical facets and statuses; no observer-level quantity is applicable	Limitations: Taxonomy facets are engineering classifications and remain provisional where the literature supports competing accounts. Boundary: Catalog coverage is bounded to this registered package and does not enumerate all known illusions. Accessibility: Every status and facet is written as text; color only reinforces the printed status.

Generated audit tables II

fig:visual panel; canonical stimulus

output/data/visual panel.json;
gregory1997visual,
brugger1999duckrabbit,
howe2005muller,
morgan1999poggendorff,
fisher1967ponzo,
yildiz2022ponzo,
kanizsa1976contours,
mruczek2015ebbinghaus,
earle1995zollner,
wertheimer1912motion,
sekuler1996wertheimer

Controls: one default typed parameter object per implemented visual entry; seed $s=0$; dimensions, luminance bounds, grayscale, and quantization controls remain in each generator schema; temporal entries display their first frame but retain sequence metrics Objective: per tile: mean and standard-deviation relative luminance, unique raster levels, and canonical SHA-256 digest; temporal entries additionally retain frame count, frame rate, and frame deltas; definitions and units are in the source data JSON

Limitations: The gallery is complete for implemented visual entries in this package, not a complete survey of visual illusions; literature families, historical displays, and DuckRabbit rasters are not pixel-identical by default. Viewing scale, display calibration, viewing distance, and observer conditions can change the relevance of a construction; the gallery does not measure an observer's perceptual report. Boundary: The figure establishes coverage of the package's implemented visual constructions and their deterministic raster facts. It does not establish that any viewer will perceive the named signature, nor that the set is exhaustive of the visual-illusion literature. Accessibility: Each tile is labeled by stable illusion ID and accompanied by printed luminance, level-count, and digest fields; color only reinforces grouping and is not required to identify a stimulus.

Generated audit tables III

fig:visual sweep; physical metric

output/data/visual sweep.json;
brugger1999duckrabbit,
gregory1997visual

Controls: duck weight {0,.25,.5,.75,1}; grayscale=quantization {2,4,8,16,32}; seed s=0
Objective: 25 canonical rasters; unique levels, normalized mean luminance, and effective dynamic range per cell

Limitations: The grid is a sensitivity of the construction parameters, not a psychophysical sensitivity curve or a validated observer effect.
Boundary: The chosen grid is an engineering sampling of the parameter domain and does not imply an optimal or perceptually uniform spacing.
Accessibility: Row and column labels state the parameter values; numerical metrics and units are available in the sidebar.
Limitations: Frame succession and frame deltas are physical file properties; playback timing, display persistence, and motion reports require a controlled observer task. Boundary: The figure documents temporal structure rather than the presence, direction, or strength of perceived motion. Accessibility: Frame index, seconds, frames per second, and normalized-difference labels remain interpretable without color.

fig:temporal sequence; physical metric

output/data/temporal sequence.json;
wertheimer1912motion, sekuler1996wertheimer

Controls: visual.apparent motion default typed parameters; frame rate and frame count from the canonical sequence; seed s=0
Objective: timestamps in seconds, frame rate in frames/s, frame count, rational timebase, and adjacent-frame normalized pixel difference

Generated audit tables IV

fig:audio signals; physical metric

output/data/audio signals.json; shepard1984scale, zatorre2005missing, repp1997tritone, deutsch1986tritone, deutsch1974octave, warren1970continuity, riecke2011continuity

Controls: default typed audio parameters; per-generator channel layout and sample rate; seed $s=0$; one-sided FFT display with 48 sampled bins Objective: RMS and peak in normalized amplitude; spectral centroid and bandwidth in Hz; sample rate in samples/s; channel layout and canonical SHA-256 digest

Limitations: Playback level, transducer response, dichotic separation, masker design, and listener judgments are external to the canonical buffer. Boundary: The figure compares generated signal properties and literature-linked families; it does not establish pitch, continuity, or stream organization for a listener. Accessibility: Waveform, spectral, amplitude, and frequency labels are printed; bar color is not the sole encoding.

fig:audiovisual timeline; physical metric

output/data/audiovisual timeline.json; shams2000sifi, hirst2020sound, vroomen2004temporal, hartcherobrien2011temporal

Controls: audiovisual.sound induced flash default shared timeline; declared sync offset ΔAV and spatial offset; seed $s=0$ Objective: video luminance and audio envelope on normalized shared time; ΔAV in ms; spatial discrepancy in normalized units; frame and sample clocks in the sidecar

Limitations: Display refresh, audio hardware, event latency, and observer temporal binding are not measured by this figure. Boundary: A shared-clock construction is not evidence that observers bind the events at the declared or any other perceptual time. Accessibility: The two channels, time axis, signed offset convention, and normalized spatial discrepancy are labeled in text.

Generated audit tables V

fig:encoding verification;
encoded media

output/data/encoding
verification.json;
engineering:typed media
adapters

Controls: typed encoding
profile selected per artifact;
overwrite and backend
capability checks; seed
inherited from the stimulus
request Objective: format
availability and verification
scope are environment
observations; NPZ is exact
while other formats are
decoded and
tolerance-checked

Limitations: Backend
availability, codec versions,
container metadata, and
lossy tolerances are
environment-dependent;
playback software can
expose additional behavior.
Boundary: A successful
encode or decode does not
validate an observer effect
or guarantee cross-device
perceptual equivalence.
Accessibility: Format,
backend, preserved fact,
and verification columns are
textual and remain usable
without color.

fig:synthetic psychophysics;
synthetic model output

output/data/synthetic
psychophysics.json;
observer:preregistered
estimands

Controls: duck weight
comparison against fixed
reference; serialized feature
weights and temperature;
seed s=0 Objective: model
probability and score delta
on the displayed
model-output scale;
canonical stimulus digests;
human data=false; training
data=none

Limitations: The model is
hand-specified, has no
training data, has not been
calibrated against observers,
and does not estimate a
human threshold or effect
size. Its probabilities are
model outputs, not
participant responses or a
psychometric function.
Boundary: This diagnostic
tests deterministic
orchestration and
metamorphic input-output
behavior only; empirical
psychophysics remains
future work. Accessibility:
Model scale, reference line,
model identity, and human d

Generated audit tables VI

fig:claim boundary; source supported
output/data/claim boundary.json;
formalism:eq:observer estimand,
formalism:eq:canonical digest

Controls: claim levels in the taxonomy and manifest; seed not applicable to the explanatory diagram
Objective: no unit-bearing objective quantity is applicable; stage definitions and evidence boundaries are preserved in the sidecar

Limitations: The diagram is a documentation contract and does not replace a preregistered observer study or empirical data.

Boundary: A deterministic artifact can support a future hypothesis but cannot supply the observer data required to test it.

Accessibility: Each stage is named, numbered, and described in text; the boundary rule is readable without color.

Limitations: The offline snapshot validates citation and lineage structure; live resolver status is a separate audit, and classifications can remain contested.

Boundary: Scholarship coverage bounds the package's evidence record and does not establish that any generated stimulus produces a universal percept. Accessibility: Source roles, claims, limitations, and statuses are printed or available in the sidecar; no category is encoded by color alone.

fig:scholarship map; source supported
output/data/scholarship map.json;
gregory1997visual, brugger1999duckrabbit, yildiz2022ponzo, hirst2020sound, bruns2019ventriloquist

Controls: checked-in evidence matrix; source roles and entry statuses; audit date recorded in data/evidence matrix.json
Objective: one evidence row per catalog entry (18 entries); source-tier counts, exact claim text, engineering basis, limitation, and gap status

Generated audit tables VII

fig:formalism traceability; physical metric	output/data/formalism traceability.json; formalism:eq:canonical digest, formalism:eq:clock definition, formalism:eq:objective statistics	Controls: formalism registry version; seed not applicable to the traceability diagram Objective: nine equation records with implementation, test, figure, and claim-level fields; no unit-bearing measurement is plotted	Limitations: Traceability records documentation and verification scope; it does not establish the truth of an observer-level theory. Boundary: A linked equation and test can show an implemented contract, not a validated perceptual law. Accessibility: Equation labels, code paths, tests, and figure labels are written as text.
fig:metrics dashboard; physical metric	output/data/metrics dashboard.json; formalism:eq:objective statistics, formalism:eq:temporal spectral metrics	Controls: default canonical artifacts; metric computation version and tolerance recorded in the source-data sidecar; seed s=0 Objective: finite objective values with units: normalized luminance/amplitude, Hz, normalized pixel difference, ms, counts, and durations	Limitations: Metrics are media properties and do not encode pitch, size, motion, localization, binding, or other observer interpretations. Boundary: Metric reproducibility is a package property; interpretation as perception requires a separate observer design and data. Accessibility: Every numerical value is paired with a printed unit or an explicit dimensionless definition.
fig:parameter domains; canonical stimulus	output/data/parameter domains.json; formalism:eq:typed request	Controls: parameter schema version and validated scalar domains; seed not applicable to the domain diagram Objective: dimensionless, pixel, level, Hz, samples/s, frames/s.	Limitations: The intervals are software validation bounds and do not encode safe listening levels, display limits, or psychophysical thresholds. Boundary: A valid parameter is a

Generated audit tables VIII

fig:observer protocol;
observer hypothesis

output/data/observer
protocol.json;
observer:preregistered
estimands

Controls: study design and deterministic randomization seed recorded in the sidecar; synthetic response generation is separate from human data Objective: trial counts, condition structure, estimand templates, response schemas, and model families; participant outcomes are not applicable

Limitations: The scaffold specifies future data collection and analysis but supplies no participant responses, fitted coefficients, power claim, or validated observer effect. Boundary: The protocol is ready for ethical and preregistered extension, not evidence that the proposed effect exists. Accessibility: Every protocol step and the no-participant-data boundary is stated in text; arrows and labels remain legible without color.

Generated audit tables IX

Table 2: Source-tiered evidence, DOI/URL verification, exact supported claims, engineering departures, limitations, and explicit planned/input-required gaps.

Key	Record and citations	DOI and URL	Verification	Supported claim / engineering / limitation
gregory 1997 visual	review; gregory 1997 visual	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	A non-exhaustive classification of visual illusion families. A review of sound-induced flash paradigms and multisensory temporal inference. A statistical image-source account of Müller-Lyer geometry. A mechanistic account of Poggendorff orientation estimation. A primary investigation of identical targets embedded in an angular context. Canonical illusory-contour inducer arrangements. Context-dependent size judgments in Ebbinghaus-family stimuli.
hirst 2020 sound	review; hirst 2020 sound	DOI: recorded; URL host/path: www.sciencedirect.com	snapshot validated; offline snapshot	
howe 2005 muller	theory; howe 2005 muller	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
morgan 1999 poggendorff	theory; morgan 1999 poggendorff	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
fisher 1967 ponzo	primary; fisher 1967 ponzo	DOI: recorded; URL host/path: www.nature.com	snapshot validated; offline snapshot	
kanizsa 1976 contours	primary; kanizsa 1976 contours	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
mruczek 2015 ebbinghaus	primary; mruczek 2015 ebbinghaus	DOI: recorded; URL host/path: pmc.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	

Generated audit tables X

wertheimer 1912 motion	primary; wertheimer 1912 motion	DOI: not recorded; URL host/path: bibbase.org	snapshot validated; offline snapshot	Foundational apparent-motion experiments using successive spatial events.
sekuler 1996 wertheimer	review; sekuler 1996 wertheimer	DOI: recorded; URL host/path: journals.sagepub.com	snapshot validated; offline snapshot	A review connecting Wertheimer's successive-event findings to later apparent-motion research and clarifying their historical scope.
shepard 1984 scale	primary; shepard 1984 scale	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	Auditory scale and pitch-class organization in Shepard-like tones.
zatorre 2005 missing	review; zatorre 2005 missing	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	The missing-fundamental problem as a pitch-inference phenomenon.
deutsch 1986 tritone	primary; deutsch 1986 tritone	DOI: recorded; URL host/path: online.ucpress.edu	snapshot validated; offline snapshot	The tritone paradox and dependence on pitch-class context and listener.
deutsch 1974 octave	primary; deutsch 1974 octave	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	Dichotic alternating high/low tone construction underlying the octave illusion.
shams 2000 sifi	primary; shams 2000 sifi	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	The one-flash/two-beep sound-induced flash contrast.

Generated audit tables XI

bruns 2019 ventriloquist	review; bruns 2019 ventriloquist	DOI: recorded; URL host/path: pmc.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	Spatial audiovisual capture and multisensory integration constraints. Sound timing can alter the apparent temporal structure of a visual event. Speech-dependent audiovisual categorical integration, requiring validated fixtures. Auditory continuity/restoration as a future calibrated masker family. Variation in duck/rabbit ambiguity across figure variants and observers. Competing depth-based and non-depth-based explanations of Ponzo-like illusions. Listener- and context-dependent effects of spectral envelope and pitch-class structure in tritone judgments. Audiovisual timing can shift temporal judgments in a purely temporal context without spatial
vroomen 2004 temporal	primary; vroomen 2004 temporal	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
mcgurk 1976 speech	primary; mcgurk 1976 speech	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
warren 1970 continuity	primary; warren 1970 continuity	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	
brugger 1999 duckrabbit	primary; brugger 1999 duckrabbit	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
yildiz 2022 ponzo	review; yildiz 2022 ponzo	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
repp 1997 tritone	primary; repp 1997 tritone	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	
hartcherobrien 2011 temporal	primary; hartcherobrien 2011 temporal	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	

Generated audit tables XII

earle 1995 zollner	primary; earle 1995 zollner	DOI: recorded; URL host/path: journals.sagepub.com	snapshot validated; offline snapshot	Crossing oblique and long-line geometry used to measure orientation interactions in the Zöllner family.
zoellner 1860 pseudoscopy	primary; zoellner 1860 pseudoscopy	DOI: recorded; URL host/path: onlinelibrary.wiley.com	snapshot validated; offline snapshot	The nineteenth-century primary description of the crossing-line orientation family later known as the Zöllner illusion.
riecke 2011 continuity	primary; rieke 2011 continuity	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	Interrupted sounds with masking context and the separation of sensory and decisional contributions.
wagemans 2012 gestalt	review; wagemans 2012 gestalt	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	Contemporary perceptual-grouping, contour-completion, and figure-ground accounts relevant to Kanizsa-type constructions.
weintraub 1979 ebbinghaus	primary; weintraub 1979 ebbinghaus	DOI: recorded; URL host/path: pubmed.ncbi.nlm.nih.gov	snapshot validated; offline snapshot	A classic contextual-size experiment varying surround geometry and comparison conditions in Ebbinghaus-family displays.

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noppeney 2018 causal	review; noppeney 2018 causal	DOI: recorded; URL host/path: doi.org	snapshot validated; offline snapshot	Causal-inference and temporal-prediction accounts of audiovisual integration and segregation.
alhazen 1989 optics	primary; alhazen 1989 optics	DOI: not recorded; URL host/path: www.cca.qc.ca	snapshot validated; offline snapshot	The English translation and commentary preserves Ibn al-Haytham's eleventh-century Arabic optics as a historical source on direct vision and geometrical image formation.
mozi 2023 canons	review; mozi 2023 canons	DOI: recorded; URL host/path: link.springer.com	snapshot validated; offline snapshot	A modern translation and commentary documents Mohist Canon sections on knowledge, optics, and mechanics, providing a Chinese Warring States-period context for early technical reasoning about vision.
ptolemy 1996 optics	primary; ptolemy 1996 optics	DOI: not recorded; URL host/path: sites.dlib.nyu.edu	snapshot validated; offline snapshot	The scholarly translation makes Ptolemy's second-century Optics available as an early work on visual perception and mathematical visual theory.

Generated audit tables XIV

berkeley 1709 vision	primary; berkeley 1709 vision	DOI: not recorded; URL host/path: www.maths.tcd.ie	snapshot validated; offline snapshot	Berkeley's 1709 essay treats visual distance and space as mediated by learned associations rather than as a simple direct readout of visual geometry. The digitized record anchors Kircher's 1646 <i>Ars magna lucis et umbrae</i> as an early-modern work on light, shadow, and optical display. The historical account documents Molyneux's 1688 question separating tactile learning from visual recognition and its long cross-sensory afterlife. The digitized Philosophical Transactions record preserves Cheselden's 1728 case report as a historical observation relevant to visual access and the separation of observer evidence from stimulus description.
kircher 1646 light	primary; kircher 1646 light	DOI: not recorded; URL host/path: collections.st-andrews.ac.uk	snapshot validated; offline snapshot	
molyneux 2020 problem	review; molyneux 2020 problem	DOI: not recorded; URL host/path: plato.stanford.edu	snapshot validated; offline snapshot	
cheselden 1728 sight	primary; cheselden 1728 sight	DOI: not recorded; URL host/path: archive.org	snapshot validated; offline snapshot	

Generated audit tables XV

dai 2015 chinesescience	review; dai 2015 chinesescience	DOI: recorded; URL host/path: link.springer.com	snapshot validated; offline snapshot	A history of Chinese science and technology surveys ancient Chinese physics and records optical phenomena as part of a non-European history of visual science.
entry / visual.duck rabbit	engineering / limitation / gap; brugger 1999 duckrabbit; gregory 1997 visual	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The entry instantiates a controllable ambiguous-figure stimulus family. Engineering basis: Parameterized ambiguous silhouette; not a pixel-identical reproduction of a historical plate. Limitation: The generator verifies geometry and luminance, not bistable reports. Missing contract: none
entry / visual.simultaneous contrast	engineering / limitation / gap; gregory 1997 visual	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The physical center patches are matched while surround luminance differs. Engineering basis: Equal central patches on unequal luminance surrounds. Limitation: Display calibration and observer

Generated audit tables XVI

entry / visual.apparent motion	engineering / limitation / gap; wertheimer 1912 motion; sekuler 1996 wertheimer	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The sequence contains controlled successive spatial events. Engineering basis: Alternating rectangular masks at a fixed rational frame rate. Limitation: The artifact establishes temporal succession, not perceived motion. Missing contract: none
entry / audio.shepard tone	engineering / limitation / gap; shepard 1984 scale	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The audio contains octave-related partials with controlled sweep parameters. Engineering basis: Additive octave-spaced partials with a deterministic envelope. Limitation: Playback transducers and listener pitch judgments are external. Missing contract: none
entry / audio.missing fundamental	engineering / limitation / gap; zatorre 2005 missing	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The nominal fundamental component is absent from the canonical spectrum. Engineering basis: Harmonic components are synthesized while the nominal

Generated audit tables XVII

entry / audiovisual.sound induced flash	engineering / limitation / gap; shams 2000 sifi; hirst 2020 sound	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The declared beep/flash event counts and offsets are physically encoded. Engineering basis: One flash paired with one or two timed beeps on a shared clock. Limitation: The stimulus does not establish a reported flash count. Missing contract: none
entry / audiovi- sual.ventriloquist	engineering / limitation / gap; bruns 2019 ventriloquist; noppenev 2018 causal	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The audio and visual channels carry a declared spatial discrepancy. Engineering basis: Stereo panning and visual displacement encode a spatial discrepancy. Limitation: Perceived localization depends on room, headphones, display, and observer. Missing contract: none
entry / visual.muller lyer	engineering / limitation / gap; gregory 1997 visual; howe 2005 muller	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The two bar lengths are equal in the canonical raster while wing geometry varies. Engineering basis: Equal bars with controlled arrow-wing geometr. Limitation:

Generated audit tables XVIII

entry / visual.poggendorff	engineering / limitation / gap; gregory 1997 visual; morgan 1999 poggendorff	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The occluder and diagonal continuation are generated from explicit geometry. Engineering basis: A diagonal continuation is separated by a rectangular occluder. Limitation: Alignment judgments and orientation filters are not measured. Missing contract: none
entry / visual.ponzo	engineering / limitation / gap; fisher 1967 ponzo; yildiz 2022 ponzo	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	Target bars are physically equal and rails converge toward a vanishing region. Engineering basis: Equal target bars are placed inside converging rails. Limitation: Perspective interpretation is observer- and display-dependent. Missing contract: none
entry / visual.kanizsa triangle	engineering / limitation / gap; kanizsa 1976 contours; wagemans 2012 gestalt	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The image contains incomplete inducers with no explicitly drawn triangle edge. Engineering basis: Three incomplete circular inducers are

Generated audit tables XIX

entry / visual.ebbinghaus	engineering / limitation / gap; mruczek 2015 ebbinghaus; weintraub 1979 ebbinghaus	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	Central target geometry is held equal while contextual circle geometry differs. Engineering basis: Equal central targets are surrounded by different context-circle sizes. Limitation: Perceived size depends on viewing scale and context. Missing contract: none
entry / visual.zollner	engineering / limitation / gap; zoellner 1860 pseudoscopy; earle 1995 zollner; gregory 1997 visual	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The entry instantiates a controlled crossing-line orientation stimulus family. Engineering basis: Vertical long lines crossed by short oblique inducers with explicit normalized spacing and angle. Limitation: The raster verifies line geometry, not orientation judgments or a pixel-identical historical reproduction. Missing contract: none
entry / audio.tritone paradox	engineering / limitation / gap; deutsch 1986 tritone; repp 1997 tritone	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The canonical pair has an explicit half-octave frequency relation. Engineering basis:

Generated audit tables XX

entry / audio.octave illusion	engineering / limitation / gap; deutsch 1974 octave	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The two channels receive alternating octave-related tones. Engineering basis: Alternating high and low tones are assigned to opposite stereo channels. Limitation: The auditory percept depends on dichotic presentation and listener. Missing contract: none
entry / audio.auditory continuity	engineering / limitation / gap; warren 1970 continuity; riecke 2011 continuity	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The canonical audio contains a reproducible interruption interval and masker family. Engineering basis: An interrupted sinusoidal carrier is replaced during a declared gap by a deterministic tone or white-noise masker with typed amplitude. Limitation: The generator reports the physical interruption and masker, not a listener's continuity judgment or sensory- versus-decisional effect. Missing contract: none

Generated audit tables XXI

entry / audiovisual.temporal ventriloquism	engineering / limitation / gap; vroomen 2004 temporal; hartcherobrien 2011 temporal; hirst 2020 sound; noppenev 2018 causal	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	Audio and video event timing and declared offset are deterministic and inspectable. Engineering basis: A visual flash and audio click share a clock with explicit offset. Limitation: The generated single-event pair is a simplified engineering stimulus. Missing contract: none
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Generated audit tables XXII

entry / audiovisual.mcgurk	engineering / limitation / gap; mcgurk 1976 speech	DOI: n/a; URL host/path: data/evidence_matrix.json	snapshot validated; offline snapshot	The catalog identifies a speech-dependent audiovisual family without claiming implementation. Engineering basis: Requires validated speech audio/video fixtures and licensing records. Limitation: No fixture, consent record, or perceptual validation contract is bundled. Missing contract: A licensed or consented checksummed speech audio/video fixture, synchronization contract, and preregistered perceptual validation protocol are required.
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Generated audit tables XXIII

Table 3: Formal equation-to-code traceability.

Label	Definition and symbols	Implementation and tests	Figures and claim level
eq:typed request	$q = (i, s, e)$ Symbols: i, s, e	Code: registry.py, schema.py; Tests: test generators.py	Figures: fig:architecture, fig:formalism traceability; Level: canonical stimulus
eq:canonical generation	$A = G(; s)$ Symbols: A, G, s	Code: registry.py, generators.py; Tests: test generators.py, test v03	Figures: fig:architecture, fig:visual panel; Level: canonical stimulus
eq:canonical digest	$c(A) = \text{Serialize_LE, float32}(A, \text{shape, clock, units}); h_c(A) = H(c(A))$ Symbols: $c(A), h_c, H$	Code: canonical.py, manifest.py; Tests: test v03 contracts.py	Figures: fig:architecture, fig:encoding verification; Level: physical metric
eq:encoding verification	$F = E(A), I = D(F), V(F, M) \in \{\text{pass, fail}\}$ Symbols: F, E, D, I, V, M	Code: render.py, inspection.py, manifest.py; Tests: test media and render.py, test v03 contracts.py	Figures: fig:encoding verification; Level: encoded media
eq:clock definition	$N = \text{round}(f_s T), t_k = k/f_r, \Delta_AV = t_audio - t_video$ Symbols: $N, f_s, T, t_k, f_r, \Delta_AV$	Code: parameters.py, artifacts.py; Tests: test parameters.py, test media and render.py	Figures: fig:temporal sequence, fig:audiovisual timeline; Level: physical metric
eq:objective statistics	$_Y, _Y, RMS_X = M(A; \text{units, tolerance, version})$ Symbols: $_Y, _Y, RMS_X, M$	Code: metrics.py, publication.py; Tests: test v04 scholarly.py	Figures: fig:metrics dashboard, fig:audio signals; Level: physical metric
eq:temporal spectral metrics	$D_k = \text{mean absolute difference of adjacent frames}; \text{centroid}(X) = \sum fP_X(f)/\sum P_X(f)$ Symbols: D_k, P_X, f	Code: metrics.py, artifacts.py; Tests: test v04 scholarly.py	Figures: fig:temporal sequence, fig:metrics dashboard; Level: physical metric

Generated audit tables XXIV

eq:observer estimand	$= E[Y \text{ conditional on condition and protocol}]$ Symbols: θ, Y, condition, protocol	Code: observer analysis.py, observer.py; Tests: test v04 scholarly.py	Figures: fig:synthetic psychophysics, fig:observer protocol; Level: observer hypothesis
eq:synthetic observer	$p_M(c \text{ given } A_r, A_c) = \frac{((w \cdot (A_c) - w \cdot (A_r)) /)}{p_M, c, A_r, A_c, w, \theta}$ Symbols:	Code: synthetic psychophysics.py, metrics.py; Tests: test synthetic psychophysics.py	Figures: fig:synthetic psychophysics; Level: synthetic model output

Generated audit tables XXV

Table 4: Claim ledger separating code-derived facts, scholarship, engineering departures, limitations, scope, and observer hypotheses.

Claim ID	Claim and level	Basis	Lineage and limitation
catalog:count	The catalog contains 18 entries. Level: canonical_stimulus	derived_from_code	Lineage: taxonomy entries() Limitation: Catalog membership is not a completeness claim about all known illusions. Manuscript: manuscript/09 appendix catalog.md
evidence:source count	The checked-in evidence matrix contains 36 source records. Level: source_supported	checked_in_scholarship	Lineage: data/evidence matrix.json::sources Limitation: The offline snapshot does not imply that every URL is currently reachable or that a source supports more than its exact record. Manuscript: manuscript/06 scope and related work.md
scope:no participant data	No participant data are bundled with the package. Level: observer_hypothesis	derived_from_code	Lineage: experiments/observer protocol.md; output/data/synthetic psychophysics.json Limitation: Future observer studies require preregistration, consent, calibrated presentation, and separate data governance. Manuscript: manuscript/07 publication audit.md

Generated audit tables XXVI

synthetic:diagnostic boundary	The synthetic observer is a deterministic model-output diagnostic, not human psychophysics. Level: synthetic_model_output	synthetic_model_output	Lineage: src/duckrabbit/synthetic psychophysics.py; output/data/synthetic psychophysics.json Limitation: The hand-specified model has no training data and has not been calibrated against observers. Manuscript: manuscript/02 methodology.md
cover:editorial boundary	The cover is a publication illustration, not an experimental stimulus or observer result. Level: publication_illustration	publication_illustration	Lineage: output/reports/cover visualization.json Limitation: The editorial asset is not part of the deterministic scientific stimulus registry. Manuscript: manuscript/07 publication audit.md
evidence:visual.duck rabbit	The entry instantiates a controllable ambiguous-figure stimulus family. Level: source_supported	checked_in_scholarship	Lineage: sources=brugger1999duckrabbit; gregory1997visual; data/evidence matrix.json; engineering=Parameterized ambiguous silhouette; not a pixel-identical reproduction of a historical plate.; limitation=The generator verifies geometry and luminance, not bistable reports.; missing contract=none Limitation: The generator verifies geometry and luminance,

Generated audit tables XXVII

evidence:visual.simultaneous contrast The physical center patches are matched while surround luminance differs. Level: source_supported checked_in_scholarship

Lineage:
sources=gregory1997visual;
data/evidence matrix.json;
engineering=Equal central patches on unequal luminance surrounds.;
limitation=Display calibration and observer adaptation are not modeled.; missing contract=none Limitation: Display calibration and observer adaptation are not modeled. Manuscript: manuscript/06 scope and related work.md

evidence:visual.apparent motion The sequence contains controlled successive spatial events. Level: source_supported checked_in_scholarship

Lineage:
sources=wertheimer1912motion;
sekuler1996wertheimer;
data/evidence matrix.json;
engineering=Alternating rectangular masks at a fixed rational frame rate.;
limitation=The artifact establishes temporal succession, not perceived motion.; missing contract=none Limitation: The artifact establishes temporal succession, not perceived motion. Manuscript: manuscript/06 scope and related work.md

Generated audit tables XXVIII

evidence:audio.shepard tone	The audio contains octave-related partials with controlled sweep parameters. Level: source_supported	checked_in_scholarship	Lineage: sources=shepard1984scale; data/evidence matrix.json; engineering=Additive octave-spaced partials with a deterministic envelope.; limitation=Playback transducers and listener pitch judgments are external.; missing contract=none Limitation: Playback transducers and listener pitch judgments are external. Manuscript: manuscript/06 scope and related work.md
evidence:audio.missing fundamental	The nominal fundamental component is absent from the canonical spectrum. Level: source_supported	checked_in_scholarship	Lineage: sources=zatorre2005missing; data/evidence matrix.json; engineering=Harmonic components are synthesized while the nominal fundamental is omitted.; limitation=Pitch completion is an observer-level inference.; missing contract=none Limitation: Pitch completion is an observer-level inference. Manuscript: manuscript/06 scope and related work.md
evidence:audiovisual.sound induced flash	The declared beep/flash event counts and offsets are physically encoded. Level: source_supported	checked_in_scholarship	Lineage: sources=shams2000sifi; hirst2020sound; data/evidence matrix.json; engineering=One flash

Generated audit tables XXIX

evidence:audiovisual.ventriloquist
The audio and visual channels carry a declared spatial discrepancy. Level: source_supported

checked_in_scholarship

Lineage:
sources=bruns2019ventriloquist;
noppenny2018causal;
data/evidence matrix.json;
engineering=Stereo panning and visual displacement encode a spatial discrepancy.;
limitation=Perceived localization depends on room, headphones, display, and observer.; missing contract=none Limitation: Perceived localization depends on room, headphones, display, and observer. Manuscript: manuscript/06 scope and related work.md

evidence:visual.muller lyer
The two bar lengths are equal in the canonical raster while wing geometry varies. Level: source_supported

checked_in_scholarship

Lineage:
sources=gregory1997visual;
howe2005muller;
data/evidence matrix.json;
engineering=Equal bars with controlled arrow-wing geometry.; limitation=The chosen normalized geometry is one engineering variant.; missing contract=none Limitation: The chosen normalized geometry is one engineering variant. Manuscript: manuscript/06 scope and related work.md

Generated audit tables XXX

evidence:visual.poggendorff	The occluder and diagonal continuation are generated from explicit geometry. Level: source_supported	checked_in_scholarship	Lineage: sources=gregory1997visual; morgan1999poggendorff; data/evidence matrix.json; engineering=A diagonal continuation is separated by a rectangular occluder.; limitation=Alignment judgments and orientation filters are not measured.; missing contract=none Limitation: Alignment judgments and orientation filters are not measured. Manuscript: manuscript/06 scope and related work.md Lineage: sources=fisher1967ponzo; yildiz2022ponzo; data/evidence matrix.json; engineering=Equal target bars are placed inside converging rails.; limitation=Perspective interpretation is observer- and display-dependent.; missing contract=none Limitation: Perspective interpretation is observer- and display-dependent. Manuscript: manuscript/06 scope and related work.md Lineage: sources=kanizsa1976contours; wagemans2012gestalt; data/evidence matrix.json;
evidence:visual.ponzo	Target bars are physically equal and rails converge toward a vanishing region. Level: source_supported	checked_in_scholarship	
evidence:visual.kanizsa triangle	The image contains incomplete inducers with no explicitly drawn triangle edge. Level:	checked_in_scholarship	

Generated audit tables XXXI

evidence:visual.ebbinghaus	Central target geometry is held equal while contextual circle geometry differs. Level: source_supported	checked_in_scholarship	Lineage: sources=mruczek2015ebbinghaus; weintraub1979ebbinghaus; data/evidence matrix.json; engineering=Equal central targets are surrounded by different context-circle sizes.; limitation=Perceived size depends on viewing scale and context.; missing contract=none Limitation: Perceived size depends on viewing scale and context. Manuscript: manuscript/06 scope and related work.md Lineage: sources=zoellner1860pseudoscopy; earle1995zollner; gregory1997visual; data/evidence matrix.json; engineering=Vertical long lines crossed by short oblique inducers with explicit normalized spacing and angle.; limitation=The raster verifies line geometry, not orientation judgments or a pixel-identical historical reproduction.; missing contract=none Limitation: The raster verifies line geometry, not orientation judgments or a pixel-identical historical reproduction. Manuscript: manuscript/06 scope and
evidence:visual.zollner	The entry instantiates a controlled crossing-line orientation stimulus family. Level: source_supported	checked_in_scholarship	

Generated audit tables XXXII

evidence:audio.tritone
paradox

The canonical pair has an
explicit half-octave
frequency relation. Level:
source_supported

checked_in_scholarship

Lineage:
sources=deutsch1986tritone;
repp1997tritone;
data/evidence matrix.json;
engineering=Two
octave-complex tones are
separated by a half-octave
interval.; limita-
tion=Ascending/descending
reports vary across listeners
and contexts.; missing
contract=none Limitation:
Ascending/descending
reports vary across listeners
and contexts. Manuscript:
manuscript/06 scope and
related work.md

evidence:audio.octave
illusion

The two channels receive
alternating octave-related
tones. Level:
source_supported

checked_in_scholarship

Lineage:
sources=deutsch1974octave;
data/evidence matrix.json;
engineering=Alternating
high and low tones are
assigned to opposite stereo
channels.; limitation=The
auditory percept depends on
dichotic presentation and
listener.; missing
contract=none Limitation:
The auditory percept
depends on dichotic
presentation and listener.
Manuscript: manuscript/06
scope and related work.md

Generated audit tables XXXIII

evidence:audio.auditory
continuity

The canonical audio
contains a reproducible
interruption interval and
masker family. Level:
source_supported

checked_in_scholarship

Lineage:
sources=warren1970continuity;
riecke2011continuity;
data/evidence matrix.json;
engineering=An interrupted
sinusoidal carrier is replaced
during a declared gap by a
deterministic tone or
white-noise masker with
typed amplitude.;
limitation=The generator
reports the physical
interruption and masker,
not a listener's continuity
judgment or
sensory-versus-decisional
effect.; missing
contract=none Limitation:
The generator reports the
physical interruption and
masker, not a listener's
continuity judgment or
sensory-versus-decisional
effect. Manuscript:
manuscript/06 scope and
related work.md
Lineage:
sources=vroomen2004temporal;
hartchero-
brien2011temporal;
hirst2020sound;
noppene2018causal;
data/evidence matrix.json;
engineering=A visual flash
and audio click share a
clock with explicit offset.;

evidence:audiovisual.temporal
ventriloquism

Audio and video event
timing and declared offset
are deterministic and
inspectable. Level:
source_supported

checked_in_scholarship

Generated audit tables XXXIV

evidence:audiovisual.mcgurk	The catalog identifies a speech-dependent audiovisual family without claiming implementation. Level: source_supported	checked_in_scholarship	Lineage: sources=mcgurk1976speech; data/evidence matrix.json; engineering=Requires validated speech audio/video fixtures and licensing records.; limitation=No fixture, consent record, or perceptual validation contract is bundled.; missing contract=A licensed or consented checksummed speech audio/video fixture, synchronization contract, and preregistered perceptual validation protocol are required. Limitation: No fixture, consent record, or perceptual validation contract is bundled. Manuscript: manuscript/06 scope and related work.md Lineage: publication caption specs() Limitation: The count is a package-state fact, not evidence of empirical validity. Manuscript: manuscript/07 publication audit.md
publication:figure count	The publication atlas contains 15 registered scientific figures. Level: physical_metric	derived_from_code	

Generated audit tables XXXV

publication:table count	The publication appendix contains the generated table set defined by the publication table payload registry. Level: physical_metric	derived_from_code	Lineage: publication table payloads() Limitation: Generated tables summarize package state and do not replace source or observer evidence. Manuscript: manuscript/07 publication audit.md
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Generated audit tables XXXVI

The source-tiered evidence table [`@tbl:evidence_audit`], caption audit [`@tbl:caption_audit`], formalism registry [`@tbl:formalism`], and claim ledger [`@tbl:claim_ledger`] are generated sidecars. They provide a publication appendix without duplicating mutable counts or silently converting literature statements into package results.

The publication artifacts have two distinct provenance boundaries. Scientific figures are deterministic PNG renders whose source-data JSON and SHA-256 digests are checked against the code-owned registry. The cover is an editorial charcoal illustration: its selected candidate, prompt fingerprint, source asset digest, dimensions, variants, and provenance boundary are recorded in `output/reports/cover_visualization.json`. Neither boundary is an observer result, and neither substitutes for display calibration or behavioral data.

Generated audit tables XXXVII

The publication audit additionally resolves every formalism edge to a real package module, test file, manuscript equation label, and registered figure. Figure source-data sidecars use a versioned schema with figure identity, generator, seed, and nested data payload; the registry verifies this identity and hash independently of the renderer. This makes a visually plausible but stale or relabeled figure fail the same provenance gate as a tampered stimulus.